Example The expected value of the voltage across a resistor is 50 V. However, the measurement gives a value of 49 V. Find:

a) Absolute error

b) % error

c) Relative accuracy

d) % accuracy

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Solution

a) $e = X_t - X_m = 50V - 49V = 1V$

b) $\%Error = \left|\frac{X_t - X_m}{X_t}\right| \times 100\% = \left|\frac{50V - 49V}{50V}\right| \times 100\% = 2\%$

c) $A = 1 - \%Error = 1 - 2\% = 0.98$

d) $\%Acc = 100\% - 2\% = 98\%$

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Example The expected value of the current through a resistor is 20 mA. However, the measurement yields a current value of 18 mA. Calculate:

a) Absolute error

b) % error

c) Relative accuracy

d) % accuracy

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Solution

a) $e = X_t - X_m = 20 - 18 = 2\ mA$

b) $\%Error = \left|\frac{X_t - X_m}{X_t}\right| \times 100\% = 10\%$

c) $A = 1 - \%Error = 1 - 0.1 = 0.90$

d) $\%\ Acc = 100\% - 10\% = 90\ \%$

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Example The accuracy of five digital voltmeters are checked by using each of them to measure a standard 1.0000V from a calibration instrument.

The voltmeter readings are as follows:

$V_1 = 1.001$ V,

$V_2 = 1.002$,

$V_3 = 0.999$,

$V_4 = 0.998$, and

$V_5 = 1.000$.

Calculate the average measured voltage, the average deviation, and the standard deviation.

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Solution

$$V_{av} = \frac{V_1 + V_2 + V_3 + V_4 + V_5}{5}$$

$$= \frac{1.001 + 1.002 + 0.999 + 0.998 + 1.000}{5} = 1.000V$$

$$d_1 = V_1 - V_{av} = 1.001 - 1.000 = 0.001V \quad \& \quad d_2 = V_2 - V_{av} = 1.002 - 1.000 = 0.002V$$

$$d_3 = V_3 - V_{av} = 0.999 - 1.000 = -0.001V \quad \& \quad d_4 = V_4 - V_{av} = 0.998 - 1.000 = -0.002V$$

$$d_5 = V_5 - V_{av} = 1.000 - 1.000 = 0V$$

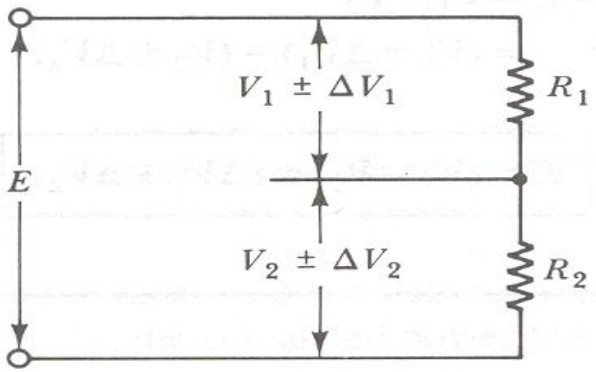
$$D = \frac{|d_1| + |d_2| + \ldots + |d_5|}{5} = \frac{0.001 + 0.002 + 0.001 + 0.002 + 0}{5} = 0.0012V$$

$$\sigma = \sqrt{\frac{d_1^2 + d_2^2 + \ldots + d_5^2}{5}} = \sqrt{\frac{(0.001)^2 + (0.002)^2 + (0.001)^2 + (0.002)^2 + 0}{5}} = 0.0014V$$

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Measurement Error Combinations

- **Sum of Quantities**
  - When a quantity is determined as the sum of two measurements, the total error is the sum of the absolute errors in each measurement.



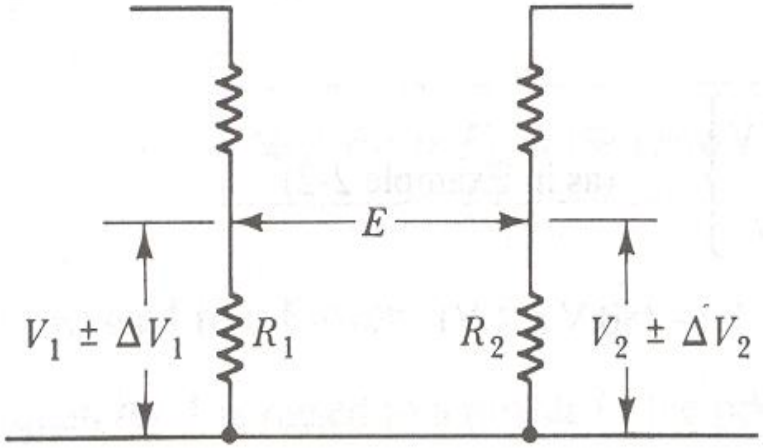
$$E = V_1 + V_2$$
$$= (V_1 \pm \Delta V_1) + (V_2 \pm \Delta V_2)$$
$$= (V_1 + V_2) \pm (\Delta V_1 + V_2)$$

(a) Error in sum of quantities equals sum of errors

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Measurement Error Combinations

- **Difference of Quantities**
  - The error of the difference of two measurements are again additive



$$E = V_1 - V_2$$
$$= (V_1 \pm \Delta V_1) - (V_2 \pm \Delta V_2)$$
$$= (V_1 - V_2) \pm (\Delta V_1 + \Delta V_2)$$

(b) Error in difference of quantities equals sum of error

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Measurement Error Combinations

- **Product of Quantities**
  - When a calculated quantity is the product of two or more quantities, the percentage error is the sum of the percentage errors in each quantity.

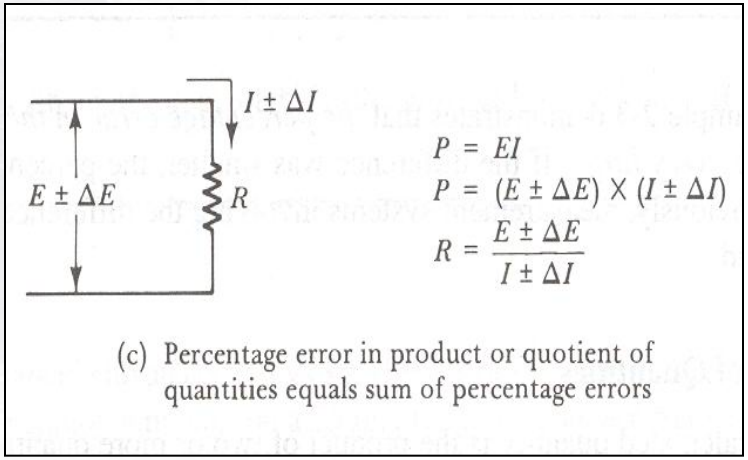
$$P = EI$$
$$= (E \pm \Delta E)(I \pm \Delta I)$$
$$= EI \pm E\Delta I \pm I\Delta E \pm \Delta E\Delta I$$

since $\Delta E\,\Delta I$ is very small,

$$P \approx EI \pm (E\Delta I \pm I\Delta E)$$

$$\text{percentage error} = \frac{E\,\Delta I + I\,\Delta E}{EI} \times 100\%$$

$$= \left(\frac{\Delta I}{I}\right) + \left(\frac{\Delta E}{E}\right) \times 100\%$$



(c) Percentage error in product or quotient of quantities equals sum of percentage errors

$$\%\text{ error in } P = (\%\text{ error in } I) + (\%\text{ error in } E)$$

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Measurement Error Combinations

- **Quotient of Quantities**

$$\%\ error\ \ in\ \frac{E}{I} = \left(\%\ error\ \ in\ E\right) + \left(\%\ error\ \ in\ I\right)$$

- **Quantity Raised to a Power**

$$\%\ error\ \ in\ A^B\ = B\left(\%\ error\ \ in\ A\right)$$

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Measurement Error Combinations

Example: An 820Ω resistance with an accuracy of ±10% carries a current of 10 mA. The current was measured by an analog ammeter on a 25mA range with an accuracy of ±2% of full scale. Calculate the power dissipated in the resistor, and determine the accuracy of the result.

Solution:

$$P = I^2 R = (10\,mA)^2 \times 820\Omega$$
$$= 82\,mW$$

$$error\ in\ R = \pm 10\%$$
$$error\ in\ I = \pm 2\%\ of\ 25\,mA$$
$$= \pm 0.5\,mA$$
$$= \frac{\pm 0.5\,mA}{10\,mA} \times 100\% = \pm 5\%$$

$$\%error\ in\ I^2 = 2(\pm 5\%) = \pm 10\%$$
$$\%error\ in\ P = (\%error\ in\ I^2) + (\%error\ in\ R)$$
$$= \pm(10\% + 10\%) = \pm 20\%$$

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